Provakar Mondol

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Research Interest

- Deep Learning
- Medical Imaging
- Image Processing
- Remote Sensing
- Video Processing
- Bioinformatics

Education

- **Khulna University of Engineering and Technology (KUET)** Khulna, Bangladesh
  M.Sc. *in Electrical and Electronic Engineering; CGPA: 3.83/4.00* July 2021 - Ongoing
  Thesis: *Optimized Deep Learning Models for Gastrointestinal Image Classification and Automated Polyp Detection*
- **Khulna University of Engineering and Technology (KUET)** Khulna, Bangladesh
  B.Sc. *in Electrical and Electronic Engineering; CGPA: 3.52/4.00* March 2013 – May 2017
  Last 2 years (81.25 credits) CGPA: 3.77
  Thesis: *Design of A New Dry Electrode and Comparison of Performance with Biopac Wet Electrode*

Research Experience

- **Gastrointestinal Image Classification and Automated Polyp Detection (2023-Present)**
  - **Multiclass GI image classification** with customized CNNs, which resulted in an accuracy of 87.41% and other bettered metrics to support endoscopic inspection.
  - **Colorectal polyp segmentation** using U-Net + Watershed with an improved IoU of 82.21%.
  - **Real-time polyp detection** from colonoscopy videos using YOLO-based models.
  - **Published 2 IEEE international conference papers** on this work.
- **Framework for Diagnosis of Alzheimer's Disease in Data-Constrained Scenarios (2023-Present)**
  - **MRI-based Alzheimer's classification** using augmentation, SMOTE and WGAN-GP synthetic data generation to address class imbalances and scarcity of data, achieving 99.19% accuracy.
  - **Transfer learning with three state-of-the-art CNNs** for performance benchmarking.
  - These works resulted in the publication of **2 IEEE international conference papers.**
- **Development of A Dry Electrode for Bio Signal Acquisition (2016-2017)**
  - **Developed a novel dry electrode** for biosignal acquisition (ECG, EEG, EMG).
  - **Benchmarked performance** against Biopac wet electrodes for validation.
- **Short Channel Effects Suppression in a Dual-Gate Gate-All-Around Si Nanowire Devices (2015-2016)**
  - **Designed a dual-gate GAA silicon nanowire nMOSFET** and compared short-channel parameters with conventional GAA.
  - Contributed to **2 IEEE international conference publications** based on this research.

Publications

- **P. Mondol**, P. Das, and K. K. Halder, "U-Net and Watershed Based Deep Learning Approach for Panoptic Segmentation of Colorectal Polyps in Colonoscopy," IEEE International Conference on Electrical, Computer and Communication Engineering (ECCE), Chittagong, Bangladesh, 2025. [Read Here]
- **P. Mondol**, P. Das, and K. K. Halder, "GINet: A Deep Learning Approach to Gastrointestinal Image Classification," IEEE International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN), Rangpur, Bangladesh, 2025. [Read Here]
- P. Das, M. Islam, and **P. Mondol**, "Utilizing a Multi-Class CNN Model for Precise Diagnosis of Alzheimer's Disease," IEEE Global Conference on Cognitive Computing and Communication Technology (GC4T), Pune, India, 2025.
- P. Das, M. Islam, and **P. Mondol**, "A Comparative Analysis of Pre-trained Models for Identifying Alzheimer's Disease," IEEE International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN), Rangpur, Bangladesh, 2025. [Read Here]
- M. W. Rony, **P. Mondol**, and H. R. Myler, "Sensitivity of a 10nm dual-gate GAA Si Nanowire nMOSFET to Process Variation," 19th International Conference on Computer and Information

Technology (ICCIT), Dhaka, Bangladesh, 2016. [Read Here]
- M. W. Rony, P. Bhowmik, H. R. Myler, and **P. Mondol**, "Short Channel Effects Suppression in a Dual-gate Gate-All-Around Si Nanowire Junctionless nMOSFET," 9th International Conference on Electrical and Computer Engineering (ICECE), Dhaka, Bangladesh, 2016. [Read Here]

Technical & Personal Skills

- **Programming Languages & Frameworks**: Python, MATLAB, C, C++, Keras, TensorFlow
- **Hardware & Circuit Design**: Arduino, Atmel AVR, Proteus
- **Graphics Programming and Industrial Applications:** Siemens SPPA T3000, MaxDNA, Valmet VMS, Honeywell PLC, Schneider PLC, Emerson PLC, ABB PLC

Professional Experiences

- **Bangladesh India Friendship Power Company Limited** — Bagerhat, Bangladesh
  Deputy Manager, Department of Control & Instrumentation — December 2020 – Present
  Responsibilities: Leading a team of 36 engineers in tasks like control loop tuning, GUI implementation, interlock checking, and network maintenance in a 1320 MW coal power plant. Completed **procurement of goods and services worth approximately USD 2,100,000.00**.
- **Daffodil International University (DIU)** — Dhaka, Bangladesh
  Lecturer, Department of EEE — October 2017 – November 2020
  Responsibilities: Taught courses and labs on Control Systems, Electrical Machine, Electronic Devices and Circuit Theory, Electronics, Random Signals and Processes, Power System Protection, Microprocessors and Interfacing. **Supervised the theses of 6 undergraduate students**.
- **Prime University (DIU)** — Dhaka, Bangladesh
  Lecturer, Department of EEE — August 2017 – September 2017
  Responsibilities: Taught courses and labs on Control Systems, Electrical Machine, Electronic Devices and Circuit Theory to undergraduate students.

Honors and Awards (Selected)

- **Dean's Award,** KUET — for outstanding academic performance (4th year).
- **Board Scholarship (Merit-13)** - Rajshahi Board, Higher Secondary Certificate Exam (2012).
- **Board Scholarship (Merit-41)** - Rajshahi Board, Secondary School Certificate Exam (2010).

Supervision Experiences (Selected)

- **Mentor, Line-Follower Robot Competition (2016):** Supervised participants, enforced rules, and evaluated performance.
- **Core & Promotional Member, IUTF at KUET (2016):** Assisted in organizing and promoting an interuniversity tech competition platform.
- **Technical Operator, IEEE EICT Conference (2015):** Supported technical operations during an international academic conference.

Standard Test Scores

- **TOEFL iBT - 106**
  Reading- 28, Listening- 29
  Speaking- 23, Writing- 26
- **GRE – 313**
  Quant- **167**, Verbal- 146
  AWA- **4.0**

References

- **Dr. Kalyan Kumar Halder**
  Professor, Department of EEE,
  Khulna University of Engineering and Technology,
  Khulna, Bangladesh
  Email: kalyan@eee.kuet.ac.bd
  [Google Scholar]
- **Dr. Naruttam Kumar Roy**
  Professor, Department of EEE,
  Khulna University of Engineering and Technology,
  Khulna, Bangladesh
  Email: nkroy@eee.kuet.ac.bd
  [Google Scholar]
- **Dr. Apan Dastider**
  Assistant Professor,
  Engineering Technology,
  Wayne State University,
  Michigan, USA
  Email: ic4711@wayne.edu
  [Google Scholar]